INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab 06: S-boxes *March 26, 2026*

Please do the following exercises by your own using C/C++ programming language and bitwise operators. The evaluation of these exercises is as follows:

- **Evaluation in the current lab session: source code at least for exercises a) and b) and programs running: 3 points.**
- Report: 2 points.
- Quiz (April 13): 5 points

1. Programming exercises

1. Substitution is an important operation involved in the design of a block cipher. A common way to do this is using a bijective function $S : \{0,1\}^\ell \to \{0,1\}^\ell$. This function S is used to replace ℓ bits in a binary string with a different ℓ bits. For example if we want to substitute 4 bits, we need to define an $S : \{0,1\}^4 \to \{0,1\}^4$. Suppose that we define S (note that we are using hexadecimal representation) as follows:

z	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$S(z)$	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7

If we have as input 0x4352(in hexadecimal) then we substitute each block of 4 bits, using S in the following way $S(4) = 2$; $S(3) = 1$; $S(5) = F$; $S(2) = D$. Thus the output will be 0x21FD.

a) Implement a function that receives values for $\ell = 2^n$, where $n \geq 2$ and randomly generate a function $S : \{0,1\}^\ell \to \{0,1\}^\ell$. Your function must store the generated table for S in a text file.

b) Implement a function that take as input a char array M and a function $S : \{0,1\}^4 \to \{0,1\}^4$ then substitute each block of 4 bits in $M[i]$ with the four bits given by S. Print M and the result of the substitution in hexadecimal. Also print each step of the substitution process. To implement this function use bitwise operators. Repeat the process but now use a function $S : \{0,1\}^8 \to \{0,1\}^8$ to substitute the 8 bits of $M[i]$

c) Implement a function that take as input an unsigned short int K and two different functions $S_1 : \{0,1\}^8 \to \{0,1\}^8$ and $S_2 : \{0,1\}^8 \to \{0,1\}^8$ then substitute the 8 most significant bits of K with the 8 bits given by S_1 and the least significant bits of K with the 8 bits of S_2. Print

K and the result of the substitution in hexadecimal. Also print each step of the substitution process. To implement this function use bitwise operators.

d) Implement a function that take as input an unsigned short int variable $K = b_{15}b_{14}\ldots b_1b_0$ and a functions $S : \{0,1\}^4 \to \{0,1\}^4$, use S to substitute each block of four bits in K. Print K and the result of the substitution in hexadecimal. Also print each step of the substitution process. To implement this function use bitwise operators.

e) Implement a program to test your functions.

2. Examples

For each function in Section 1 *Programming exercises* write down by hand an example showing step by step how the bitwise operations change M or K

3. Products

Every student must write her/his own document to present results. Your document must include

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code you implemented for exercises b), c) and d).
3. Screenshots of your source code running.
4. Examples written by hand for Section 2 *Exercises*